

1.1 WHAT IS SYSTEMS ENGINEERING?

There are many ways in which to define systems engineering. For the purposes of this book, we will use the following definition:

The function of systems engineering is to *guide the engineering of complex systems*.

The words in this definition are used in their conventional meanings, as described further below.

To guide is defined as “to lead, manage, or direct, usually based on the superior experience in pursuing a given course” and “to show the way.” This characterization emphasizes the process of selecting the path for others to follow from among many possible courses—a primary function of systems engineering. A dictionary definition of engineering is “the application of scientific principles to practical ends; as the design, construction and operation of efficient and economical structures, equipment, and systems.” In this definition, the terms “efficient” and “economical” are particular contributions of good systems engineering.

The word “system,” as is the case with most common English words, has a very broad meaning. A frequently used definition of a system is “a set of *interrelated*

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systems engineering.

The relation of modern systems engineering to its origins can be best understood in terms of three basic factors:

1. *Advancing Technology*, which provide opportunities for increasing system capabilities, but introduces development risks that require systems engineering management; nowhere is this more evident than in the world of automation. Technology advances in human–system interfaces, robotics, and software make this particular area one of the fastest growing technologies affecting system design.
2. *Competition*, whose various forms require seeking superior (and more advanced) system solutions through the use of system-level trade-offs among alternative approaches.
3. *Specialization*, which requires the partitioning of the system into building blocks corresponding to specific product types that can be designed and built by specialists, and strict management of their interfaces and interactions.

These factors are discussed in the following paragraphs.

Consequently, a successful systems engineer has the following traits and attributes:

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- a good problem solver and should welcome challenges;
- well grounded technically, with broad interests;
- analytical and systematic, but also creative; and
- a superior communicator, with leadership skills.

required at some point in the future. It attempts to anticipate developmental problems and to resolve them as early as possible in the development cycle; where this is not practicable, it establishes contingency plans for later implementation as required.

Successful system development requires the use of a consistent, well-understood systems engineering approach within the organization, which involves the exercise of systematic and disciplined direction, with extensive planning, analysis, reviews, and documentation. Just as important, however, is a side of systems engineering that is often overlooked, namely, innovation. For a new complex system to compete successfully in

intended to illustrate the general nature of these differences. Using the three dimensions to represent technical depth, technical breadth, and management depth, respectively, it is seen that the design specialist may have limited managerial skills but has a deep populous as the material and energy entities among system functions, it is convenient to subdivide them into two classes: (1) elements dealing with propagating information (e.g., radio signals), to be referred to as *signal elements*, and (2) those dealing with stationary information (e.g., computer programs), to be referred to as *data elements*. The former class is primarily associated with sensing and communications and the latter with analysis and decision processes. This results in a total of four classes of system functional elements:

1. *Signal Elements*, which sense and communicate information;
2. *Data Elements*, which interpret, organize, and manipulate information;
3. *Material Elements*, which provide structure and transformation of materials;
and
4. *Energy Elements*, which provide energy and motive power.

To provide a context for acquainting the student with significant design knowledge peculiar to each of the four broad classes of functional elements, a set of generic func-

The source of complexity in enterprise systems engineering is primarily the integration of a diversity of systems and processes. The enterprise typically includes the following components that must be integrated together under the inherent uncertainty of today's enterprise:

- business strategy and strategic planning,
 - business processes,
 - enterprise services,
 - governance,
 - technical processes,
 - people management and interactions,
 - knowledge management,
 - information technology infrastructure and investment,
 - facility and equipment management,
 - supplies management, and
 - data and information management.
- It is a complex effort.
 - It meets an important user need.
 - It usually requires several years to complete.
 - It is made up of many interrelated tasks.

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- It involves several different disciplines.
- It is usually performed by several organizations.
- It has a specific schedule and budget.